

CLEARSKY OMEGA · COMPANION DOCUMENT

Value Stack Walkthrough

A companion to the Value Stack Funnel: the two flows drawn separately, one sample site walked through both, and what to confirm with DividendVPP and Lightsmith.

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STATUS

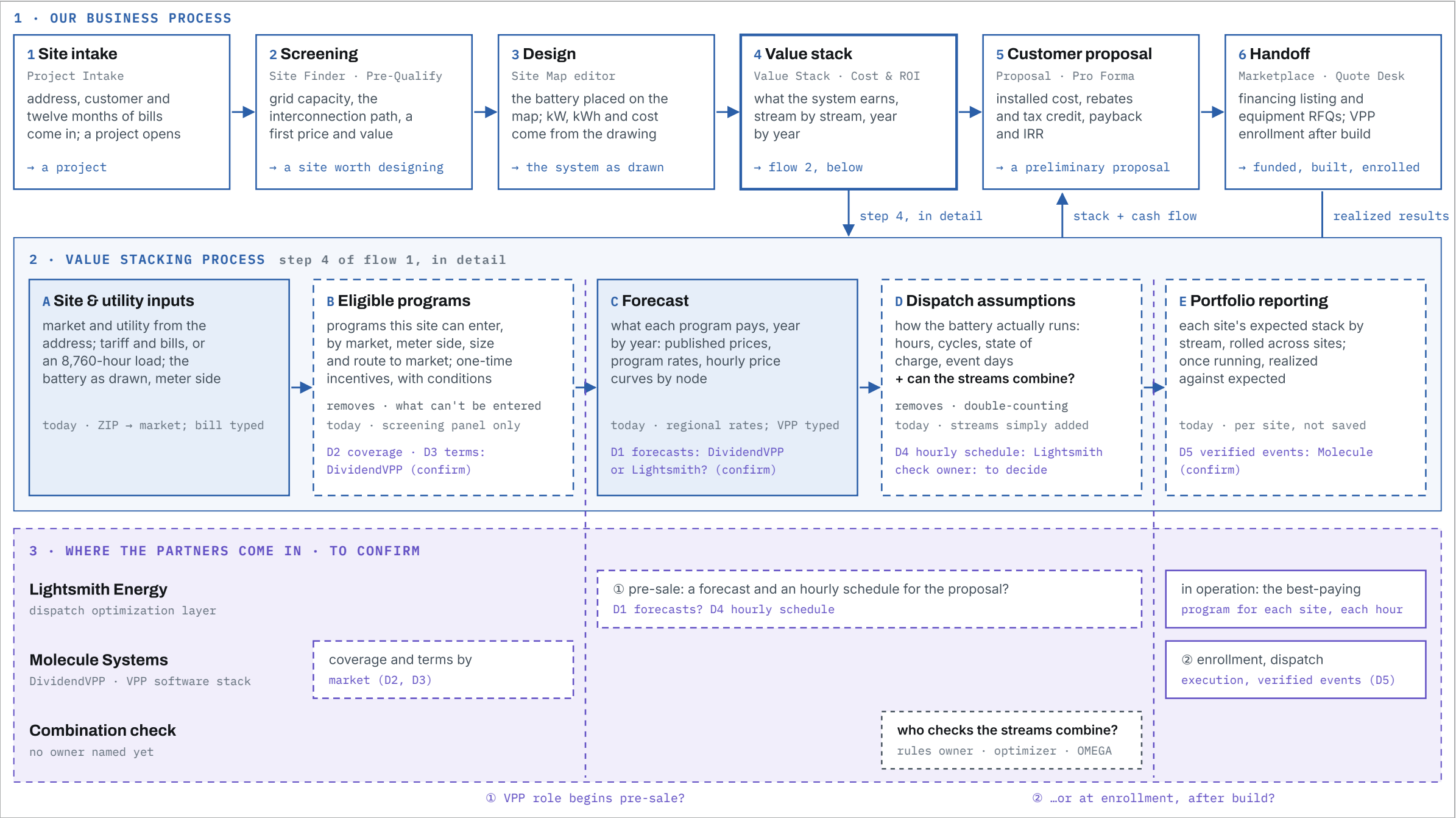
Discussion draft

DATE

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Two flows, one project. Flow 1 is our business process, from site intake and screening through the customer proposal and handoff. Flow 2 is step 4 of it: how the value stack is built, from site and utility inputs through eligible programs, the forecast and dispatch assumptions to portfolio reporting.

Today the number a customer sees rests on the inputs, a regional forecast and a ten-year cash flow. Program eligibility, dispatch and the check that revenue streams can be combined are **not yet applied to it**. That is where DividendVPP and Lightsmith come in, and where we need to agree who does what.



One sample site, through both flows

For walking the flows in the meeting. The site is illustrative; the Year-1 figures are what OMEGA's server-side screening model returns for it (the ComEd / PJM scenario), each labeled with where it comes from.

Sample site illustrative, for the walkthrough	
SITE C&I building, Chicago, IL; peak demand above 1 MW	UTILITY · MARKET ComEd · PJM, ComEd zone
METER SIDE Behind the meter	BATTERY 1,000 kW / 4,000 kWh, 4-hour
BILLS Twelve months, to replace the screening demand band	ROUTE TO MARKET To confirm in the meeting

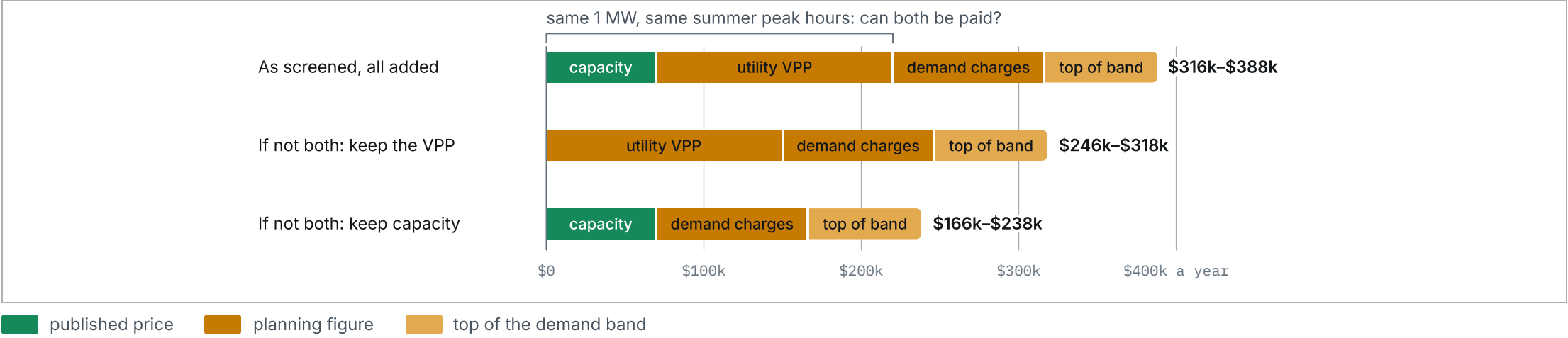


Figure 2. The sample site's Year-1 stack as OMEGA's screening model returns it, and what it becomes if PJM capacity and the utility VPP program can't both count the same 1 MW. Colors show where a figure comes from, not which stream it is.

FLOW 1, FOR THIS SITE

1 · Site intake The address, the customer and twelve months of bills come in through Project Intake; a project opens.	2 · Screening Site Finder prices it on the ComEd / PJM screening scenario; Grid Pre-Qualify checks the ComEd interconnection path.	3 · Design A 1 MW / 4 MWh battery is placed on the site map, behind the meter.	4 · Value stack Flow 2, below: the figures in the chart above.	5 · Customer proposal Installed cost, the ComEd rebate and the tax credit, payback and IRR, marked preliminary until verified.	6 · Handoff A financing marketplace listing and equipment RFQs; VPP enrollment after build.
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FLOW 2, FOR THIS SITE

STEP	WHAT GOES IN	WHAT COMES OUT
A · Inputs	ComEd, PJM (ComEd zone); behind the meter; 1,000 kW for 4 hours. Until the bills are read, the model uses a demand-charge band of \$8–14 per kW-month.	The market, meter side and size that decide which programs exist for this site.
B · Eligible programs	PJM's 4-hour storage class; the utility VPP program (ComEd's is still before the ICC); the ComEd storage rebate; the federal tax credit.	PJM capacity, accredited at 59%; demand charges avoided; the utility VPP. One-time: the rebate at \$250/kWh, which needs a paired DG facility and Rate BESH, and the 30% tax credit, which needs prevailing wage. Route to market: to confirm.
C · Forecast	PJM's 2028/29 capacity auction at \$325/MW-day; the demand-charge band; a \$150/kW-year planning figure for the VPP.	\$69,989 capacity published · \$96,000–\$168,000 demand charges planning · \$150,000 utility VPP planning . Added: \$315,989–\$387,989 a year. One-time rebate: \$1,000,000.
D · Dispatch assumptions	All three streams want the battery in the same summer peak hours, and a 4-hour battery makes one full discharge a day.	The combination check. If the same 1 MW can't earn PJM capacity and the VPP program at once, the stack falls to \$246k–\$318k or \$166k–\$238k. Nothing checks this today.
E · Portfolio reporting	The site's expected stack by stream and its ten-year cash flow.	Rolled into the portfolio; once running, realized revenue per stream reported against this expectation.

Lightsmith and DividendVPP: what to confirm

What OMEGA has on record for each partner, and what we need each to confirm: the inputs they take, the outputs they return, where their VPP role begins, and who checks that revenue streams can actually be combined.

PARTNER	Lightsmith Energy DISPATCH OPTIMIZATION LAYER	Molecule Systems · DividendVPP VPP SOFTWARE STACK
ON RECORD IN OMEGA	Cross-program optimization, so each site is in the best-paying program available each hour.	AERA integrates the devices; each dispatch event is executed and verified by telemetry. A control and optimization layer, not a curtailment provider: a route to market sits underneath it.
TAKES FROM OMEGA CONFIRM	The asset (kW, kWh, round-trip efficiency, warranty cycles), an 8,760-hour load, the tariff, the node, and the eligible programs and enrollments. No customer name or address.	The commissioned asset and telemetry access, the programs chosen, and the customer's acceptance.
RETURNS TO OMEGA CONFIRM	An hourly program schedule and dispatch plan, forecast revenue per stream, and the value forgone where streams collide.	Coverage and terms by market (D2, D3); enrollment status; verified events and realized revenue by program (D5). A price forecast, if the "DividendVPP forecast" deck type is theirs (D1).
VPP ROLE BEGINS DECIDE	① Pre-sale, so the proposal carries its forecast and schedule; or ② only once the asset is enrolled.	At enrollment, after build. Coverage and terms shape the proposal before that.
TERMS CONFIRM	None on record.	Our tools assume different shares; we need one figure per market and program.

Who checks that revenue streams can actually be combined? Today, no one. OMEGA's revenue model adds the streams, and no program's rules on combining them are held anywhere in OMEGA.	
On the sample site , PJM capacity (\$69,989) and the utility VPP (\$150,000) would both count the same 1 MW in the same summer peak hours, and managing the demand charge needs the battery at the monthly peak as well. Whether the same kW can be paid twice is a program rule, not a modeling choice. The answer moves this site's stack by \$70k to \$150k a year .	The market participant of record, the aggregator or program operator, owns each program's combination rules, because the program holds them to it. Lightsmith's optimizer enforces the rules hour by hour, choosing one program for an hour where two can't combine. OMEGA applies the rules before a stack reaches a proposal: a stream the rules exclude isn't added, and the proposal says which combination it assumes.
Proposed: the participant of record owns the rules, Lightsmith enforces them in dispatch, and OMEGA applies them in the stack. To decide in the meeting: the named owner, and where the rules table lives.	

What each point would send, and in what shape

D1 to D5 are the five points from the Value Stack Funnel, placed here on the value stacking steps. Enrollment would follow the partner route OMEGA already runs for equipment vendors: the vendor sees an anonymous opportunity (state, utility, size, stage), and the customer's identity is released only when the customer accepts.

POINT	FROM CONFIRM	WHAT	SHAPE OMEGA NEEDS	WHERE IT LANDS
D1 Rates & forecasts	DividendVPP or Lightsmith	Rates for the programs you enroll assets in; hourly price forecasts	Rate, unit (per kW-year, per kW-summer, per event), program, utility, source and effective date. Forecasts as 8,760 hourly rows per product per settlement point, unit declared (\$/MWh or \$/MW-h), negative prices kept, a vintage label.	Step C. Fills the VPP line, typed in today, and the program and capacity rates, each with its source. A forecast becomes a price deck locked to its vintage.
D2 Coverage	DividendVPP, with the route to market	Markets, utilities and programs you can enroll into	Per program: market, utility territory, behind or in front of the meter, minimum kW, telemetry, season.	Step B. Which streams the route reaches, resolved by utility territory rather than by state.
D3 Terms	DividendVPP, with the route to market	Platform fee or revenue share	Per market and program, with any minimums.	Step B. What stays with the owner on this route.
D4 Dispatch	Lightsmith	A co-optimized hourly schedule	Request: kW, kWh, round-trip efficiency, warranty cycles, 8,760-hour load, tariff, node, and no customer name or address. Response: hourly charge, discharge and state of charge; revenue per stream; value forgone.	Step D. An hourly run on one state of charge instead of streams added together: the number a lender funds.
D5 Verification	Molecule (telemetry)	A figure you stand behind; verified events once running	Verification ID, date, the assumptions behind it, and how long it holds; each event's telemetry once enrolled.	Steps D and E. Replaces the "~ Estimate" badge and travels with the project to reports, the pro forma and the financing listing.